

CHANDRA PRAKASH SHARMA

PH.D. RESEARCHER - AEROSPACE ENGINEERING - IISc BENGALURU

CONTACT

Location
Bengaluru, Karnataka 560012

LinkedIn
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CORE SKILLS

- Structural Dynamics
- Wave Propagation Analysis
- Structural Finite Element Analysis
- Guided Wave Propagation
- Spectral Finite Element Analysis
- Fracture Mechanics
- Vibration Analysis

TOOLS

- ANSYS
- MATLAB
- Python
- AutoCAD
- CATIA V5
- SolidWorks
- C++

FOCUS

- Quasicrystal-reinforced structures
- Band-gap engineering
- Dynamic response reduction
- Structural health monitoring

P RESEARCH PROFILE

Ph.D. Researcher at the Indian Institute of Science (IISc) Bengaluru, specializing in computational mechanics and advanced material modeling for aerospace structures. Research centers on spectral finite element analysis (SFEA), guided and leaky-Lamb wave propagation, and phononic band-gap engineering in periodic and quasicrystal-reinforced aluminum structures. Experienced in static, free-vibration, and wave-propagation modeling to achieve superior vibration isolation and dynamic response attenuation at the intersection of materials science and structural dynamics.

E EDUCATION

Ph.D. - Aerospace Engineering (Ongoing) 2021 - Present
Indian Institute of Science (IISc), Bengaluru | CGPA: 8.0 / 10

- Guided wave propagation, structural health monitoring, and band-gap analysis in periodic structures.
- Static and dynamic response computation using Spectral Finite Element Analysis (SFEA).

M.Tech - Machine Design (Mechanical Engineering) 2018 - 2020
IIT (ISM) Dhanbad | OGPA: 8.83 / 10

- Focused on fracture analysis and fatigue life estimation of biomedical hip implants subjected to variable loading conditions.

B.Tech - Mechanical Engineering 2013 - 2017
Institute of Engineering & Technology, Lucknow (AKTU) | 71.62%

- Worked on Thermoacoustic Refrigeration System prototype - achieved 8 deg C temperature differential.

R PUBLICATIONS

[1] Guided Wave Propagation in Quasicrystal Plate and Quasicrystal Reinforced Structures
Chandra Prakash Sharma, S. Gopalakrishnan - Manuscript

[2] Static and Free Vibration Analysis of Quasicrystal and Quasicrystal-Reinforced Plates
Chandra Prakash Sharma, S. Gopalakrishnan - Manuscript

[3] Band Gap Analysis of Periodic Aluminium–Quasicrystal Structures: Phason-Tunable Vibration Isolation
Chandra Prakash Sharma, S. Gopalakrishnan - Manuscript

[4] A Study on Guided Wave Propagation in Quasicrystal Plates
Chandra Prakash Sharma, Srinivasan Gopalakrishnan - SHM2025 Conference Paper, Published

W WORK EXPERIENCE

Engineering Research Intern - Tata Motors, Lucknow | ERC Jun - Aug 2016
Computed and benchmarked vehicle performance parameters; gained exposure to automotive testing standards, measurement protocols, and engineering documentation.

CHANDRA PRAKASH SHARMA

RESEARCH PROJECTS - LEADERSHIP - OUTREACH

RESPONSIBILITIES

- Chief Coordinator - Aerospace Research Symposium (AERES24)
- Placement Coordinator - M.Tech IIT(ISM) Dhanbad & B.Tech IET Lucknow
- Secretary - Mechanical Engineering Society, IET Lucknow
- Vice-President - SAE India Collegiate Club, IET Lucknow

EXTRACURRICULAR

- Volunteer Mentor - Karmajyoti NGO for underprivileged students
- Regular blood donor since 2015
- Captain - Mechanical Football Team
- Vice-Captain - Lawn Tennis & Cricket Teams, IET Lucknow

RECRUITER VALUE

- Strong computational modeling background
- Research-to-implementation mindset
- Finite element and vibration analysis expertise
- Leadership in academic event execution

RESEARCH & PROJECTS

Wave Propagation in Quasicrystal-Reinforced Aluminium Plates 2021 - Present

Ph.D. Research - IISc Bengaluru

- Designing novel aerospace-grade materials that exploit quasicrystal phason modes and periodic structure topology to achieve superior vibration attenuation.
- Developing models for guided Lamb waves and leaky Lamb waves in quasicrystal-reinforced composite plates.
- Applying SFEA to compute static and dynamic structural responses and predict wave dispersion characteristics.

Fracture Analysis of Hip Prosthesis under Fatigue Loading 2019 - 2020

M.Tech Thesis - IIT (ISM) Dhanbad

- Determined critical crack size of a Ti-6Al-4V alloy prosthetic hip under dynamic fatigue loading using fracture mechanics principles.
- Predicted fatigue life from crack initiation to final fracture.

Prototype - Thermoacoustic Refrigeration System 2016 - 2017

B.Tech Final Year Project - IET Lucknow

- Designed and fabricated a standing-wave thermoacoustic refrigerator; achieved an 8 deg C temperature differential across the stack element.
- Optimised stack geometry and resonator length to improve coefficient of performance.

EVENT & OUTREACH

Chief Coordinator - Aerospace Research Symposium (AERES24) 2024

IISc Bengaluru

- Organised and led a major inter-institutional aerospace research symposium, coordinating speakers, scheduling, and logistics for 150+ attendees.

SELECTED STRENGTHS

> Combines aerospace structural dynamics, finite element modeling, guided-wave physics, and advanced material design. The profile is suitable for research, R&D, computational mechanics, structural dynamics, simulation, and academic/industrial roles involving mechanics-based modeling and analysis.